European Commission

PRODUCT USER MANUAL

For Mediterranean Water Mass Formation Rates from Reanalysis

OMI_EXTREME_WMF_MEDSEA_area_averaged_mean

Issue 1.2

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Change Record

Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	16-June-2022	all	Creation of the document	Emanuela Clementi	Rita Lecci	
1.1	16-June-2023	all	Extension of the time series to year 2021	Emanuela Clementi, Pietro Miraglio	Rita Lecci	Copernicus Marine Product Management
1.2	28-May-2025	all	Temporal extension, Setup of operational extension and renaming of the product name	Rafael Gomes de Menezes	Emanuela Clementi	Copernicus Marine Product Management

GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
MED	Mediterranean
MEDREA	Mediterranean Sea (Med-MFC) Reanalysis
MFC	Monitoring and Forecasting Center
MOI	Mercator Océan International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
WMF	Water Mass Formation

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Ocean Monitoring Indicator (OMI) OMI_EXTREME_WMF_MEDSEA_area_averaged_mean: it provides time series of Water Mass Formation Rates (WMF) area averaged time series calculated in four areas of the Mediterranean Sea from 1987 to the year preceding the current one [-1Y] (including both Reanalysis and Interim data), operationally extended yearly (see the QUID document for more details on the calculation method).

It contains 1D (time dimension) WMF information computed from the Mediterranean Sea Reanalysis (Escudier et al., 2020).

OMI_EXTREME_WMF_MEDSEA_area_averaged_mean product is available in four areas characterized by dense water formation: (1) the Gulf of Lions for the Western Mediterranean Deep Waters (WMDW); (2) the Southern Adriatic Sea Pit for the Eastern Mediterranean Deep Waters (EMDW); (3) the Cretan Sea for Cretan Intermediate Waters (CIW) and Cretan Deep Waters (CDW); (4) the Rhodes Gyre, the area of formation of the so-called Levantine Intermediate Waters (LIW) and Levantine Deep Waters (LDW). Annual water mass formation rates have been computed using daily mixed layer depth estimates (density criteria $\Delta\sigma$ = 0.01 kg/m3, 10 m reference level) considering the annual maximum volume of water above mixed layer depth with potential density within or higher the specific thresholds specified in Table 1 then divided by seconds per year.

OMI_EXTREME_WMF_MEDSEA_area_averaged_mean contains annual mean time series of WMF.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

I.2. History of changes

2022/06	First delivery of the OMI
2023/06	Extension of the time series to year 2021
2025/05	Extension of the time series, Setup of operational extension and renaming of the product name

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	OMI_EXTREME_WMF_MEDSEA_area_averaged_mean
Geographical coverage	30°N, 46°N; 6°W, 36°E
Variables	Water Mass Formation (WMF)
Update frequency	Annual
Available time series	From 1987 up to one year before the current year [-1Y]
Delivery mechanism	MDS
Temporal/Spatial resolution	Annual averages, 1/24
Integration depth (for example..)	N/A
Format	NetCDF CF1.7

II.2. Details of the dataset

omi_extreme_wmf_medsea_area_averaged_mean contains the following variables:
WMDW [m3 s-1] long_name: "Gulf of Lion deep water mass formation rate" standard_name: "water_mass_formation_rate"
EMDW [m3 s-1] long_name: "South Adriatic Pit deep water mass formation rate" standard_name: " water_mass_formation_rate"
CDW [m3 s-1] long_name: "Cretan Sea deep water mass formation rate" standard_name: " water_mass_formation_rate"
LDW [m3 s-1] long_name: "Rhodes Gyre deep water mass formation rate" standard_name: " water_mass_formation_rate"
CIW [m3 s-1] long_name: "Cretan Sea intermediate water mass formation rate" standard_name: " water_mass_formation_rate"
LIW [m3 s-1] long_name: "Rhodes Gyre intermediate water mass formation rate" standard_name: " water_mass_formation_rate"

II.3. Product System Description

Data extracted from the Copernicus Marine Mediterranean Sea Reanalysis at 1/24 degree resolution provide the temperature and salinity used to evaluate the density fields which are used to compute the series of the Mediterranean WMF. The annual mean time series start on 1987, each yearly mean is centred on the 30th of June. This product includes annual mean files of WMF temporal series in four areas of the Mediterranean Sea.

III FILES NOMENCLATURE AND FORMAT

III.1. Nomenclature of files

OMI_EXTREME_WMF_MEDSEA_area_averaged_mean files nomenclature when downloaded through the CMEMS Services is based on the following nomenclature:

<datasetname >.nc

Where :

- **<datasetname>** is the dataset reference**.**
- **.nc** is the standard NetCDF filename extension.

For example: omi_extreme_wmf_medsea_area_averaged_mean.nc

III.2. File Format

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link: What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article: how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IVREFERENCES

Quality Information Document (QUID): https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-VAR-EXTREME-WMF-MEDSEA.pdf

Escudier, R., Clementi, E., Omar, M., Cipollone, A., Pistoia, J., Aydogdu, A., Drudi, M., Grandi, A., Lyubartsev, V., Lecci, R., Cretí, S., Masina, S., Coppini, G., & Pinardi, N. (2020). Mediterranean Sea Physical Reanalysis (CMEMS MED-Currents) (Version 1) [Data set]. Copernicus Monitoring Environment Marine Service (CMEMS). https://doi.org/10.25423/CMCC/MEDSEA_MULTIYEAR_PHY_006_004_E3R1